

which are now ready to be analysed to assess their risk and determine the necessary fixes.

First, let us prioritise the vulnerabilities based on the levels of severity. Click on the number of vulnerabilities (here, 19) to get the page as shown in Figure 6. This page can also be reached by clicking the main 'Scans' tab and selecting 'Results' from the dropdown menu, which typically shows an aggregated list of all the findings from all scans.

This results page (as shown in Figure 6) provides a dashboard that visually summarises all detected vulnerabilities, immediately highlighting that the majority are of Medium and Low severity with no critical threats. The problems are mostly caused by wrong settings for things like SSL/TLS and SSH (for example, using old certificates). Below the dashboard, a detailed list specifies each vulnerability, its severity score, and the affected host. The clear action plan is to use this information to prioritise and remediate the Medium-severity issues first, addressing the most significant risks before moving on to hardening the system against lower-level findings.

Common vulnerabilities and their impact

The vulnerabilities found during the scan that have an impact on the host are shown in Figure 7.

The impact of these vulnerabilities is as follows.

TCP timestamps: Exposure of TCP timestamps can lead to fingerprinting attacks, where an attacker can determine the operating system of a target system based on its TCP timestamp behaviour. This information can be used to launch more targeted attacks against systems, potentially exploiting known vulnerabilities in specific operating systems.

SSL/TLS certificate expired: Expired SSL/TLS certificates can result in insecure connections between clients

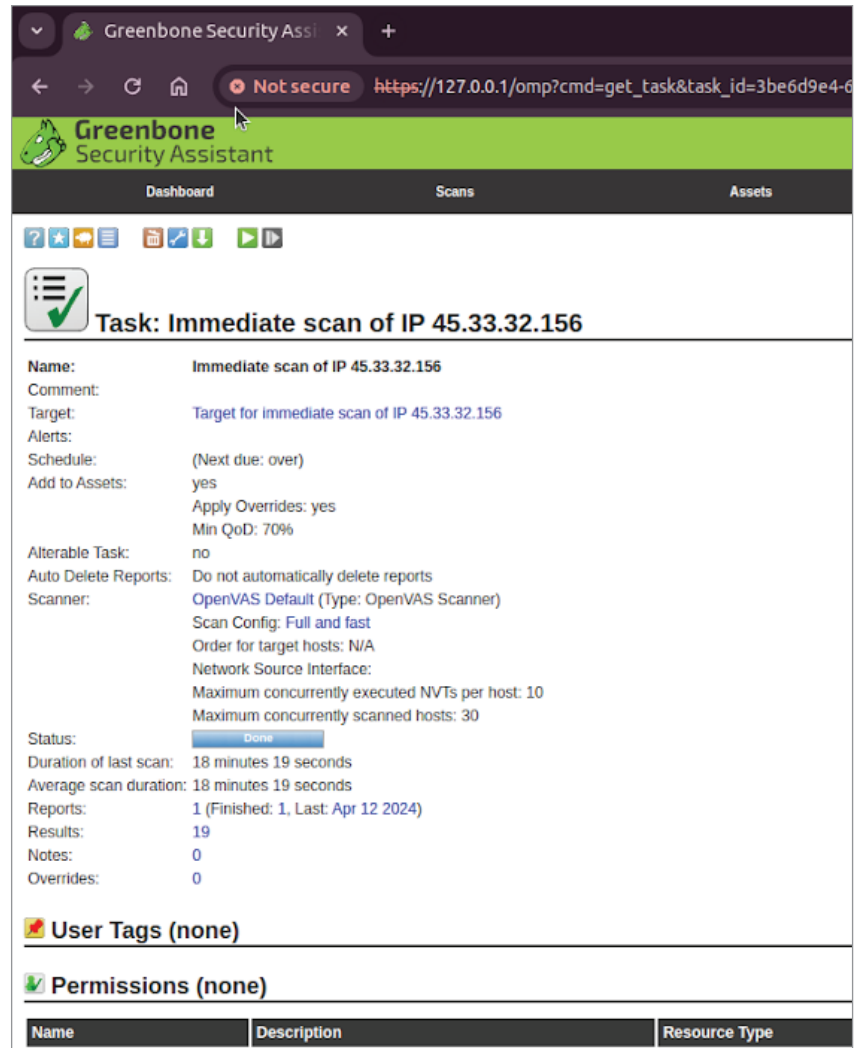


Figure 5: Task details page

and servers. Users may encounter warning messages in their browsers, potentially leading to distrust of the website. Attackers could exploit this vulnerability to intercept or manipulate communication between the client and server, leading to data breaches or other malicious activities.

SCM files accessible: Exposing source code management (SCM) files can lead to the disclosure of sensitive information such as credentials, API keys, or intellectual property. Attackers can use this information to gain unauthorised access to systems, escalate privileges, or conduct further attacks against the organisation.

SSH weak MAC algorithms supported: Weak message authentication code (MAC) algorithms can be exploited by attackers to tamper with SSH traffic, leading to data integrity violations. Attackers can potentially execute man-in-the-middle attacks, intercepting and modifying SSH communications without detection. This can lead to unauthorised access to sensitive systems and data.

SSH weak encryption algorithms supported: Weak encryption algorithms in SSH can be exploited by attackers to decrypt intercepted SSH traffic. This can lead to the exposure of sensitive information, including authentication